SUMMARY · MK 4 OPERATIONS & SUPPLY CHAIN MANAGEMENT · WEEK 4

Project Management

Projects are temporary and unique. Project types and structures, the triple constraint, WBS and milestones, CPM, three time estimates (PERT), fast tracking vs crashing and time–cost trade-offs, earned value, and choosing a delivery method. Cases: KOODESIGN and Whoosh high-speed rail.

Material

1 Projects, structures and the triple constraint

Projects are one-off, uncertain, interdependent and irreversible. Structures: pure (focus, duplication), functional (depth, low priority), matrix (both, two bosses). Triple constraint: scope, time, cost with quality absorbing squeeze — state which is fixed, firm, flexible. WBS and milestones (KOODESIGN: 19 tasks, 4 milestones).

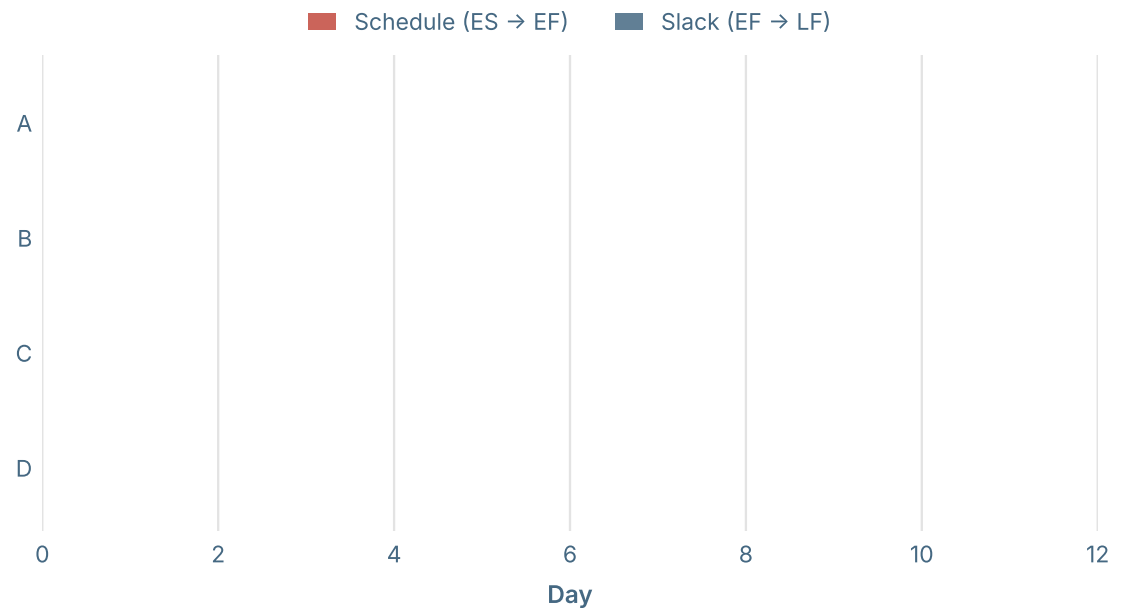
2 CPM: critical path and slack

CPM: list, sequence, forward pass (ES/EF), backward pass (LF/LS), slack = LS − ES, critical path = zero slack. Example A(3)→B(4)→D(2), A→C(6)→D: critical A-C-D = 11, B slack 2. A near-zero-slack project is fragile.

3 The sample network as a Gantt: critical path and slack made visible

Deep dive

Sample network Gantt (days)



Red = critical (zero slack).

Activity	ES	EF	LS	LF	Slack
A	0	3	0	3	0
B	3	7	5	9	2
C	3	9	3	9	0
D	9	11	9	11	0

4 Three time estimates (PERT)

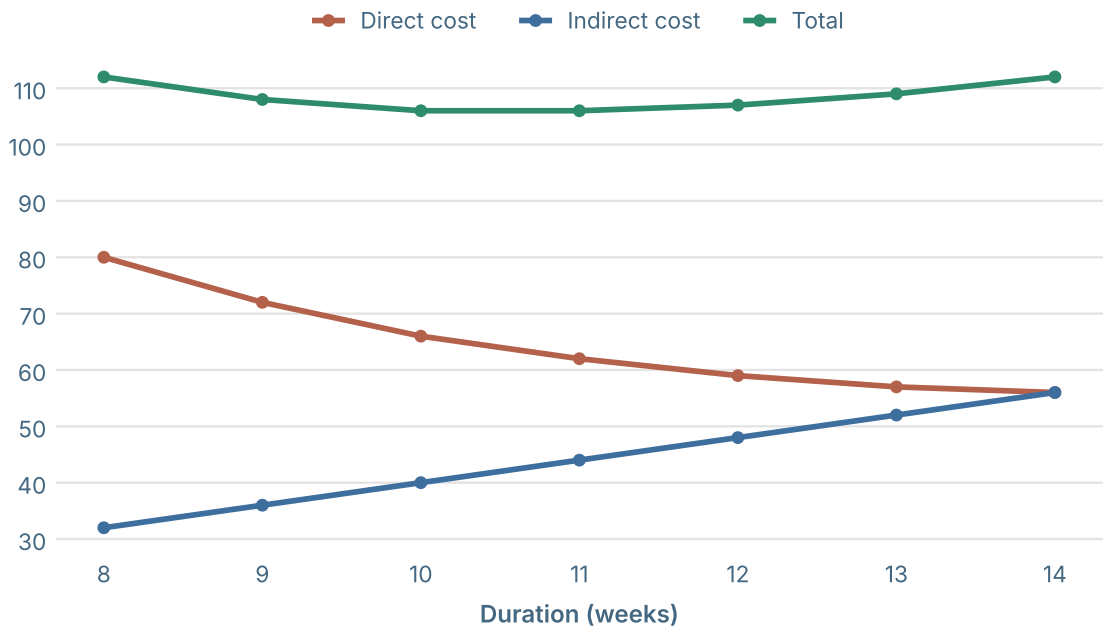
TE = (a + 4m + b)/6; variance = ((b − a)/6)². Sum along the critical path; Z = (target − TE)/σ. Example a=2, m=4, b=12 → TE 5, σ² 2.78. A 33-week path gives ~65% of finishing by week 34. State confidence, not just a date.

5 Compressing: fast tracking vs crashing

Options: reduce scope, fast track (overlap; rework risk), crash (add resources; cost), overtime, levelling, renegotiate. Crash only critical activities, cheapest slope first: (crash − normal cost)/(normal − crash time). Optimum minimises direct + indirect cost. Example: 3 weeks cut raised cost $94k → $97k.

6 The time-cost curve: why fastest ≠ cheapest Deep dive

Time-cost trade-off (illustrative)



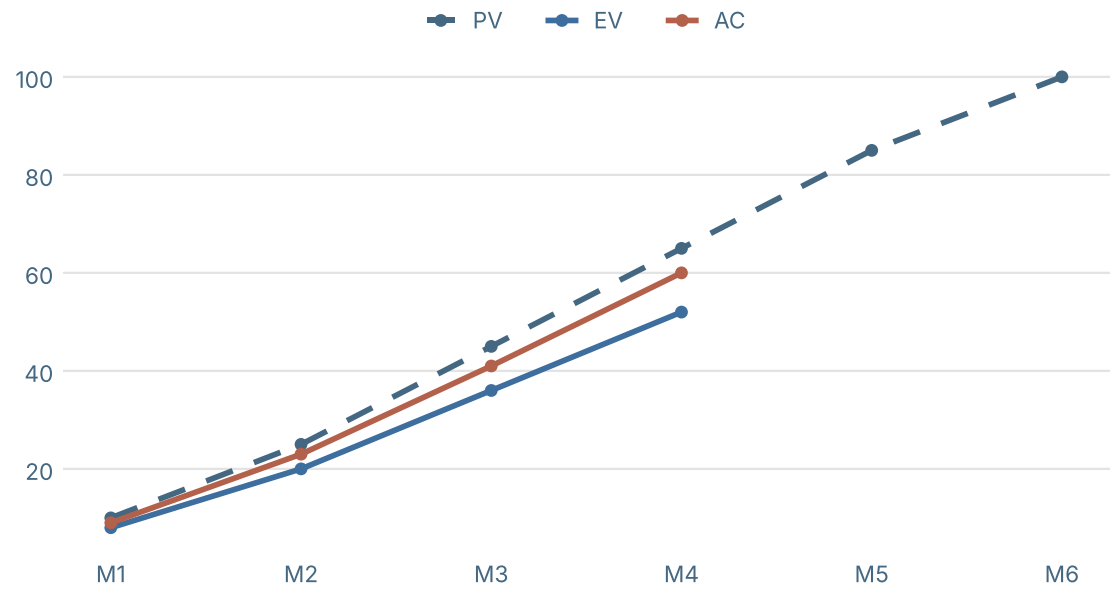
Direct cost rises as you compress; indirect cost rises as you extend. The optimum minimises the total.

7 Earned Value Management

PV, EV, AC; CV = EV − AC, CPI = EV/AC; SV = EV − PV, SPI = EV/PV; below 1 = trouble. Example PV 100, EV 80, AC 90 → CPI 0.89, SPI 0.80. EVM measures execution, not whether the plan is still worth executing.

8 Earned-value S-curves: reading a project in month 4 Deep dive

Cumulative PV, EV, AC (illustrative)



CPI = 0.87
Over budget.

SPI = 0.80
Behind schedule.

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EAC ≈ BAC ÷ CPI = 100 ÷ 0.87 ≈ 115
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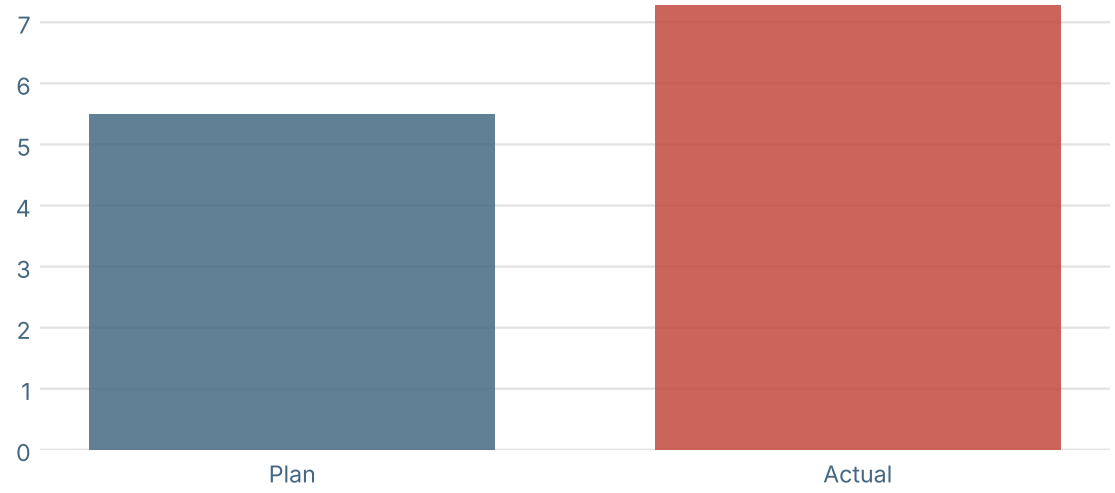
9 Choosing the delivery approach

Waterfall, Agile/Scrum, Kanban, Lean, hybrid, stage-gate, critical chain — match to uncertainty: simple → waterfall; complicated → agile/waterfall; complex → Scrum; chaotic → Kanban. Borrow iterations, WIP limits and flow visualisation while keeping milestones.

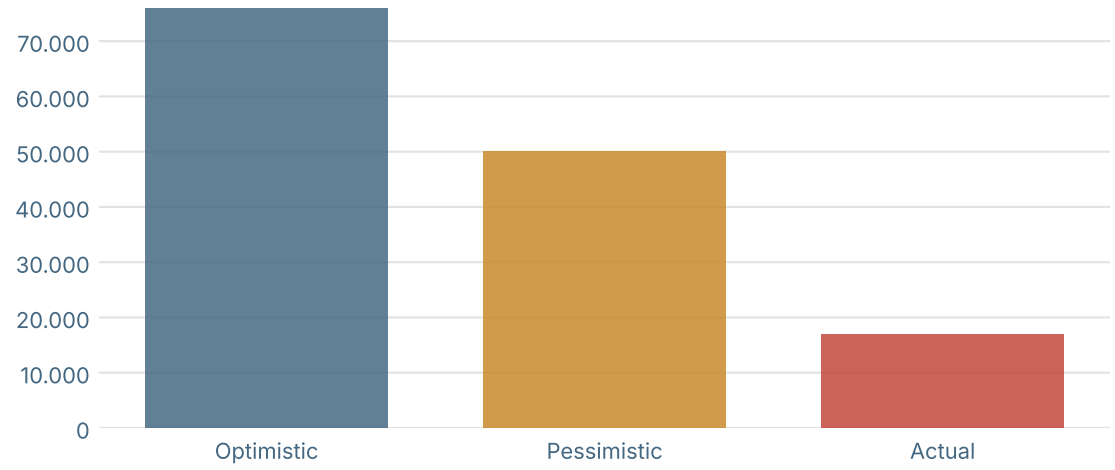
10 Whoosh in charts

Deep dive

Whoosh cost: plan vs actual (US$ bn)



Passengers per day: forecast vs actual



+31.3%
Cost overrun

4+ yrs
Late

~US$52m/km
vs 17–18m

99.9%
On time

Optimism bias

Even the pessimistic forecast was ~3× actual.

Key takeaways

1. State confidence, not just dates.
2. The triple constraint relocates pressure.
3. Critical path; slack = LS − ES.
4. TE = (a+4m+b)/6.
5. Crash the cheapest critical activity.
6. CPI = EV/AC, SPI = EV/PV.
7. Set kill criteria before investing.

Case studies

Case & worked answers

Whoosh: the project that could never be stopped

Overrun 31.3% (US$5.5bn → 7.27bn), 4+ years late, ~17,000 passengers/day vs a 50,000 "pessimistic" forecast, yet 99.9% on-time. Q1: with scope, financing and date locked, pressure moved to time and then the balance sheet. Q2: much of the ~US$1.7bn overrun is unplanned scope and duration-driven cost (land prices, interest). Q3: institutionalised optimism bias, no safe channel for bad numbers. Q4: kill metrics (cost +15%, delay >18 months, demand <60% of base) would have triggered around 2019–2021. Q5: e.g. A + D with measurable promises and self-applied kill criteria.

Worked examples

Problem 1 **Medium** **PERT dan peluang selesai tepat waktu**

Aktivitas X: optimis 4, paling mungkin 6, pesimis 14 hari. Hitung TE dan variansnya. Jika jalur kritis total TE = 20 hari dengan total varians 4, berapa peluang proyek selesai ≤ 23 hari?

1. $TE = (a + 4m + b) \div 6 = (4 + 24 + 14) \div 6 = 7$ hari
2. $\sigma^2 = ((b - a) \div 6)^2 = (10 \div 6)^2 = 2{,}78$
3. Jalur kritis: $\sigma = \sqrt{4} = 2$ hari.
4. $z = (23 - 20) \div 2 = 1{,}5 \rightarrow$ dari tabel normal, $P(z \leq 1{,}5) = 0{,}933$.

Answer: TE 7 hari, varians 2,78; peluang selesai ≤ 23 hari ± 93%.

Problem 2 Easy **Biaya crashing**

Aktivitas normal 10 hari (Rp20 juta), bisa dipercepat jadi 7 hari (Rp29 juta). Berapa biaya percepatan per hari?

1. `Biaya/hari = (crash cost − normal cost) ÷ (normal time − crash time)`
2. = (29 − 20) ÷ (10 − 7) = 9 ÷ 3 = 3.

Answer: Rp3 juta per hari. Percepat hanya jika aktivitas ini kritis dan termurah di antara aktivitas kritis.

Problem 3 Medium **Earned value**

BAC Rp500 juta. Saat ini PV = 200, EV = 180, AC = 210. Hitung CV, SV, CPI, SPI, dan EAC.

1. CV = EV − AC = 180 − 210 = −30 (boros).
2. SV = EV − PV = 180 − 200 = −20 (terlambat).
3. CPI = 180 ÷ 210 = 0,857; SPI = 180 ÷ 200 = 0,90.
4. `EAC = BAC ÷ CPI = 500 ÷ 0,857 = 583`

Answer: Over budget dan terlambat; perkiraan biaya akhir ± Rp583 juta.

Practice questions

Try answering first. The answer key is below each question.

1. Zero slack means...

A. Freely delayable
B. On the critical path
C. Finished
D. Unimportant

Answer: B. Critical path.

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2. a 2, m 4, b 12. TE = ...

A. 4
B. 5
C. 6
D. 6.5

Answer: B. 30/6.

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3. When crashing, first compress...

A. Cheapest anywhere
B. Cheapest critical activity
C. Most slack
D. All at once

Answer: B. Non-critical crashing buys nothing.

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4. EV 52, AC 60, PV 65. Status...

A. Under budget, ahead
B. Over budget, behind
C. Under budget, behind
D. Over budget, ahead

Answer: B. Both indexes below 1.

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5. With scope, financing and date locked, pressure finally landed on...

A. Technical quality
B. Time, then the balance sheet
C. Stations
D. Nothing

Answer: B. 4 years late, cost overrun.

Sources

- [FINAL] Project Management (MK4)
- Jacobs & Chase, Bab 4
- KOODESIGN case (HKUST ST131)
- Whoosh case, SBM ITB Case Series